

Chapter 7: Predictive Models

Outline

Introduction

Knowledge-based and data-driven

Optimization and parameter estimation

Model interpretation

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7.1 Introduction

- ▶ Supervised machine learning methods allow us to extract and refine *hypotheses* from data.
- ▶ These hypotheses in turn help us make predictions about new, unseen data.
- ▶ This process is called *modeling*, and we often refer to the hypotheses as *models*.
- ▶ However, modeling is a much bigger umbrella than supervised learning or even machine learning.
- ▶ In this section we expand on the concept of modeling, and how machine learning is connected to other modeling approaches.

7.1 Introduction - Definition of *Model*

- ▶ A model is a mathematical description of a physical system that can return information about a new state of the system based on the information currently available.

7.1 Introduction - Building and using models

- ▶ The level of detail in the description can vary and must be chosen.
- ▶ Detailed descriptions require complex, sophisticated models.
- ▶ Coarse descriptions require simpler models.
- ▶ In general simpler models are preferred, but they may overlook crucial aspects of the physical systems and return inaccurate information.

7.1 Introduction - Building and using models

- ▶ No model can fully capture the reality of a physical system.
- ▶ We use *assumptions* to delimit the scope of what the model is expected to capture.
- ▶ The model is expected to return valid information only if the assumptions are met.
- ▶ The more assumptions we define, the narrower the scope and simpler the model.

7.1 Introduction - Practical Definition of model

- ▶ A model is a set of rules and parameters that map a set of inputs to a set of outputs.
- ▶ In some contexts, inputs are called *independent variables* while outputs are called *dependent variables*.
- ▶ *Parameters* are quantifies that when combined with the *inputs* following the *rules* of the model produce the *outputs*.
- ▶ The degree of deviation of the model from the expected outputs is called *accuracy*.

7.1 Introduction - Practical Definition of Model

- ▶ Complexity of a model can be described by the number of rules and parameters it requires to return an output.
- ▶ The number of inputs is related to the number of physical aspects that are recognized by the model.
- ▶ The number of outputs tells us how information we can obtain about the state of the system.

7.1 Introduction - Challenges in modeling

- ▶ The task of building a model is that of finding the right set of rules and parameters.
- ▶ Measurements for inputs and outputs may not have the necessary resolution, or may not be available at all.
- ▶ The causal relationships may be too complex and *intractable*.
- ▶ In this scenario, we can simplify the model to make it tractable, but in doing so:
 - ▶ We lose accuracy.
 - ▶ Some parameters may not have physical meaning or will lump together more than one causal relationship.

7.2 Knowledge-based Modeling

- ▶ Built from in-depth understanding of the laws of physics governing the physical system.
- ▶ Consider a model that predicts the time a ball will take to reach the ground after being thrown up in the air:

$$t = v_0 \sin(\theta)/g \quad (1)$$

- ▶ This model has one rule (one equation), two inputs (initial velocity, v_0 and angle θ), and one parameter (gravitational acceleration, g) which also has physical meaning.
- ▶ This model is simple and very accurate, but tells us nothing about the properties of the ball or its surroundings. It assumes implicitly that these aspects do not affect the behavior of the system.

7.3 Data-driven Modeling

- ▶ If the laws of physics governing the system are not available, we may try to infer them from data.
- ▶ In the ball example, we can set up an experiment in which a ball is thrown up multiple times at different velocities and its position is recorder over time.
- ▶ We can then fit a mathematical function, say

$$t = v_0 A \sin(\theta) \quad (2)$$

- ▶ For a certain parameter A , our model accurately predicts the behavior of the system.
- ▶ One may be tempted to conclude, by inspection, that $A = 1/g$ but for the data driven model A is capturing not only the gravitational effects but also *all other effects* that occurred during the experiments.

7.3 Data-driven Modeling

- ▶ We can include more aspects of the system in our data driven model, using more inputs, more rules, and more parameters.
- ▶ This will take it further away from physical interpretation.
- ▶ In the extreme, our model could be able to predict the state of the system but cannot be interpreted physically at all, which limits its use.

7.4 Surrogate Modeling

- ▶ In surrogate modeling, data generated from a model is used to build another model.
- ▶ Typically, the second model is simpler, cheaper to evaluate, or more explainable.
- ▶ Surrogate modeling can be used to train ML models using knowledge-based models or the other way around, generating explicit equations models from ML model.

7.5 Hybrid Modeling

- ▶ Hybrid modeling is a broad term used to describe the combination of different models and model synthesis methods.
- ▶ This notion is fundamental in Chemical Engineering practice, where sizing, optimization, and control methods rely on the estimation of empirical parameters that are coupled into mechanistic models.
- ▶ A general but formal definition can be stated as:
 - ▶ Given a scenario description S , a domain theory Th , consisting of domain model fragments $\{m_1, \dots, m_n\}$ and a set of rules constraining their use, and a query about the scenario's behavior, obtain the most useful, coherent scenario model $M = \{m_i\sigma_i, \dots, m_k\sigma_k\}$ for answering the query.

7.5 Hybrid Modeling

- Hybrid modeling bridges knowledge-based and data driven models.

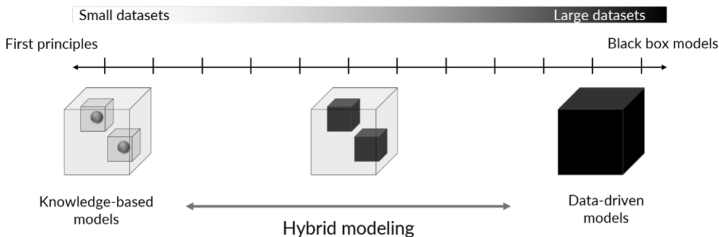


Figure: Spectrum of modeling strategies from knowledge-based to data-driven.

7.6 Search and Optimization

- ▶ The notion of search (for hypotheses, hyperparameters) introduced in previous chapters is now formalized as *optimization*.
- ▶ Given a set of functions $F = \{f_1, f_2, \dots, f_n\}$ and a set of constraints $G = \{g_1, g_2, \dots, g_m\}$ defined over a set of decision variables $x = \{x_1, x_2, \dots, x_k\}$, and optimization problem can be defined to seek the set of values of x that maximize (or minimize) the values assumed by the functions in F subject to the constraints in G .

7.6 Search and Optimization

- ▶ Optimization problems are solved using *optimization algorithms* which can be broadly classified as gradient-based or gradient-free.
 - ▶ Gradient-based use the derivatives of the functions F with respect to x to find directions of improvement.
 - ▶ Gradient-free evaluate the functions in F for many values of x to find trends and estimate directions of improvement.
- ▶ The backpropagation algorithm introduced for ANNs is an example of gradient-based optimization.

7.7 Parameter Estimation

- ▶ Backpropagation provides a way of finding the parameters W for an ANN model.
- ▶ The basic idea of using an optimization algorithm to find the parameters of a model can be extended to any model.
- ▶ This specific type of optimization problem is known as *parameter estimation*.
- ▶ Formally, for a generic function $y = f(x_1, \dots, x_n, a_1, \dots, a_m)$ where the inputs $\{x_1, \dots, x_n\}$ are mapped to y through some operations involving the coefficients (parameters) $\{a_1, \dots, a_m\}$, we want to solve the optimization problem:

$$\min_{\{a_1, \dots, a_m\}} \sum_{i=1}^N (y_i - f(x_{i1}, \dots, x_{in}, a_1, \dots, a_m))^2 \quad (3)$$

7.8 Model Interpretation and Sensitivity

- ▶ Model interpretation analyzes a model to establish the influence of input variables and/or parameters over the outputs.
- ▶ Depending on model complexity, this interpretation could be straightforward or require a more involved procedure.
- ▶ In a linear regression model, $y = mx + b$, the influence of x over y is given directly by the parameter m .
- ▶ If we try to do the same for an ANN model with, say, three layers each with three nodes, we have

$$y = \text{act}^3 (W^3 \cdot \text{act}^2 (W^2 \cdot \text{act}^1 (W^1 \cdot x + b_1) + b_2) + b_3) \quad (4)$$

- ▶ While this is still tractable, interpretation is much more complicated.

7.8 Model Interpretation and Sensitivity - Methods

- ▶ For complex models, methods like Sobol' sensitivity analysis, subspace greedy search, and model explanations, provide data-driven ways of extracting model interpretations.
- ▶ Sobol' sensitivity analysis uses the expansion of the function into orthogonal terms of increasing dimensions with mutually independent input variables:

$$f(x) = f_0 + \sum_{i=1}^n f_i(x_i) + \sum_{1 \leq i < j \leq n} f_{ij}(x_i, x_j) + \dots + f_{1,2,\dots,n}(x_1, x_2, \dots, x_n) \quad (5)$$

- ▶ Then, the variances of each orthogonal function f_i , f_{ij} , $f_{1,2,\dots,n}$ are used to calculate the sensitivity indices $\hat{S}_i = \hat{V}_i / \hat{V}$ and $\hat{S}_{Ti} = 1 - \hat{V}_{-i} / \hat{V}$.

7.8 Model Interpretation and Sensitivity - Methods

- ▶ Model explanations are procedures used to interpret ML models.
- ▶ Shapley additive explanation values (SHAP) are an example of these methods.
- ▶ SHAP values assign each input feature a contribution value and are calculated by considering all possible combinations of features and their marginal contributions to the prediction.

7.10 Conclusions

- ▶ Modeling in Chemical Engineering integrates knowledge-based and data-driven approaches, expanding the toolkit for understanding processes and phenomena.
- ▶ Surrogate and hybrid modeling were highlighted as cost-effective methods combining physical insights with data-driven techniques.
- ▶ Parameter estimation was discussed through optimization.
- ▶ Sensitivity analysis, exemplified by the Sobol' method, and a hybrid modeling case study for fuel cell design demonstrated practical applications.